

# Ignition of Ammonium Perchlorate Composite Propellants by Convective Heating

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A comprehensive experimental study was made of the ignition of ammonium perchlorate propellants in a shock-tube apparatus. Test variables were: 1) heat flux, 2) test-gas temperature, 3) test-gas velocity, and 4) pressure. Convective heat fluxes at the propellant surface were varied over the range of about 20 to 160 cal/cm<sup>2</sup> sec. The experimental results showed that, in the range of conditions used, ignition characteristics of ammonium perchlorate propellants are in agreement with those predicted by thermal ignition theory. However, for propellant samples with surface irregularities, the thermal ignition process is accelerated at low test-gas velocities by external-flux-augmenting exothermic reactions among decomposition products at the surface. These decomposition products are produced at local hot spots resulting from two-dimensional convective heating of protrusions and other surface imperfections. The experimental data indicate that the key reaction in the ignition process is the low-temperature thermal decomposition of ammonium perchlorate.

## Nomenclature

|                |                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| $B$            | = product of the frequency factor $Z$ , and the energy supplied at the propellant surface per unit area $Q_s$ by the key ignition reaction  |
| $c$            | = specific heat                                                                                                                             |
| $E_a$          | = activation energy for key ignition reaction                                                                                               |
| $F_s$          | = externally applied heat flux to propellant surface                                                                                        |
| $\bar{F}_s$    | = mean externally applied heat flux                                                                                                         |
| $\bar{F}_{sr}$ | = mean heat flux supplied at the propellant surface by secondary ignition reactions                                                         |
| $F_T$          | = total heat flux to propellant surface                                                                                                     |
| $G$            | = mass flow rate of test gas through flow channel of test section                                                                           |
| $h$            | = convective heat-transfer coefficient                                                                                                      |
| $k$            | = thermal conductivity                                                                                                                      |
| $M_{ts}$       | = Mach number of test gas flowing through test section                                                                                      |
| $P$            | = pressure                                                                                                                                  |
| $Q_s$          | = energy supplied at the propellant surface per unit area by key ignition reaction                                                          |
| $Q_v$          | = heat of reaction per unit volume of propellant                                                                                            |
| $R$            | = gas constant                                                                                                                              |
| $t$            | = time                                                                                                                                      |
| $t_i$          | = ignition time                                                                                                                             |
| $T$            | = temperature                                                                                                                               |
| $T_g$          | = stagnation temperature of convective fluid                                                                                                |
| $T_j$          | = absolute temperature at the heat flux gage or propellant surface immediately following passage of shock wave                              |
| $T_0$          | = initial uniform temperature of heat flux gage or propellant                                                                               |
| $\Delta T_0$   | = instantaneous temperature rise at the heat flux gage or propellant surface produced by the passage of the shock wave through test section |
| $T_s$          | = time-dependent surface temperature of heat flux gage                                                                                      |
| $T_{si}^L$     | = ignition temperature for linear heating of propellant surface, defined by Eq. (14)                                                        |
| $T_{si}^T$     | = thermal ignition temperature                                                                                                              |
| $x$            | = distance into the propellant, measured from its surface                                                                                   |
| $Z$            | = frequency factor for the key ignition reaction                                                                                            |

$\Gamma$  = thermal responsivity (square root of the product of  $\rho$ ,  $c$ , and  $k$ )  
 $\Gamma_p$  = propellant thermal responsivity  
 $\rho$  = density

## Introduction

IGNITION of a solid propellant is achieved through a combination of chemical and thermal processes that raise the propellant temperature and establish an environment that can support combustion reactions. To obtain useful information from ignition experiments, it is necessary to know the rate of external heat transfer to the propellant and also to probe the nature of the chemical processes important in ignition.

In recent years, several ignition theories have been proposed to explain the ignition process for propellants; however, experimental data rarely have been sufficiently reliable and complete to permit critical evaluation of the theories. One of the primary goals of the research described by this paper was to obtain reliable ignition data that could be used in such evaluation.

Foremost among the current theories on solid-propellant ignition are the following.

1) Gas-phase theory: The gas-phase theory of propellant ignition, as proposed by McAlevy and Summerfield,<sup>1,2</sup> assumes that propellant ingredients decompose by thermal processes and then undergo vigorous exothermic reaction in the gas phase, thereby transferring energy back to the propellant. The rate-controlling reactions (the slow step in the process) are presumed to be those in the gas phase near the propellant surface.

2) Thermal theory: The thermal theory of propellant ignition assumes that the rate of the ignition process is controlled by the temperature of the solid phase. For rapid heating, this would be the temperature of the propellant surface. Ignition occurs when the surface temperature of the solid reaches a level at which "boot-strapping" exothermic reactions release sensible energy at a rate exceeding the external flux. Early work on the thermal theory of ignition, as applied to solid propellants, was conducted by Frazer and Hicks,<sup>3</sup> Hicks,<sup>4</sup> and Altman and Grant.<sup>5</sup> Recent contributors to thermal ignition theory were Baer and Ryan<sup>6-8</sup> and Evans, Beyer, and McCulley.<sup>9</sup>

3) Heterogeneous-reaction theory: Anderson, Brown, and Shannon<sup>10</sup> have suggested that the critical rate-controlling reactions (the slow step) in the ignition process of ammonium

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perchlorate propellants are spontaneous heterogeneous reactions between gaseous oxidizing species produced by the decomposition of ammonium perchlorate and the solid binder-fuel.

The results of the experimental work on ammonium perchlorate propellant ignition, on which this paper is based, supplemented by earlier experimental studies at the University of Utah<sup>6-8,11</sup>, are explained best by thermal ignition theory.

## Thermal Ignition Theory

### Thermal Ignition with Bulk-Phase Reactions

Hicks<sup>4</sup> carried out a numerical integration of the equation that describes thermal ignition of a homogeneous propellant with the key ignition reaction distributed in the bulk phase:

$$\rho c(\partial T/\partial t) = k(\partial^2 T/\partial x^2) + Z Q_e e^{-E_a/RT} \quad (1)$$

at  $x = 0$ ,

$$F_s(t) = -k(\partial T/\partial x) = h(T_g - T_s)$$

at  $x = +\infty$ ,

$$\partial T/\partial x = 0 \quad \text{all } t$$

at  $t = 0$ ,

$$T(x) = T_0 \quad \text{all } x$$

The basic assumptions for this model are: 1) heat is transferred uniformly to the surface of a homogeneous, semi-infinite slab of propellant, initially at  $T_0$ , by a hot gas with a temperature ( $T_g$ ) through a constant heat-transfer coefficient ( $h$ ); 2) the propellant thermophysical properties,  $k$ ,  $\rho$ , and  $c$ , are not temperature dependent; 3) the rate of the condensed-phase reactions is not dependent on concentration of ingredients; and 4) there is no regression of the propellant surface or depletion of reactive propellant ingredients during ignition.

Evans, Beyer, and McCulley<sup>9</sup> considered a similar thermal ignition model in their study of the ignition of ammonium perchlorate-copper chromite-carbon pellets by high, radiant heat fluxes.

### Thermal Ignition with Surface Reactions

From the work of Hicks, Baer<sup>6</sup> showed that data for propellants that ignite thermally should be represented by a straight line on a plot of mean surface heat flux  $\bar{F}_s$  vs the square root of ignition time  $(t_i)^{1/2}$ , on a logarithmic plot and that the slope of the straight line that represents the experimental data is related to the activation energy of the key ignition reaction. However, it was found that Hicks' ignition model gave activation energies for the key ignition reaction which were considerably higher than those expected for the kinds of reactions which would appear to be important in ignition. Baer<sup>8</sup> suggested that the ignition process for ammonium perchlorate propellants could be better represented by a model that considered a surface reaction to be the key chemical reaction in ignition. The partial differential equation that describes thermal ignition with the key reaction localized at the surface is:

$$\rho c(\partial T/\partial t) = k(\partial^2 T/\partial x^2) \quad (2)$$

at  $x = 0$ ,

$$F_T(t) = -k(\partial T/\partial x) = F_s + Be^{-E_a/RT}$$

at  $x = +\infty$ ,

$$T(t) = T_0 \quad \text{all } t$$

at  $t = 0$ ,

$$T(x) = T_0 \quad \text{all } x$$

Equation (2) was put into a dimensionless form, and numerical solutions were obtained for a reasonable range of values of the dimensionless parameters. The following assumptions were made:

- 1) The propellant heated by a constant surface heat flux can be treated as a homogeneous, semi-infinite solid originally at a uniform temperature ( $T_0$ ).
- 2) The thermophysical properties of the propellant,  $k$ ,  $\rho$ , and  $c$ , are not temperature dependent. Values determined at 60°C were used for experimental purposes.
- 3) Negligible surface regression of the propellant surface occurs up to the time of ignition.
- 4) The rate of the surface reaction is not dependent on concentration of ingredients.
- 5) Activation energy  $E_a$  and heat of reaction  $Q_s$  are not temperature dependent.
- 6) Ignition of the propellant occurs when the heat flux contributed by the surface reaction is much greater than the applied surface heat flux.

Baer found that the solution of Eq. (2) can be represented by

$$(t_i^*)^{1/2} = [(\Pi)^{1/2}/2\bar{F}^*] \{ [1/1 - 1.04 \ln(\bar{F}^*)] - T_0^* \} \quad (3)$$

where the dimensionless parameters are

$$t_i^* = t_i(RB/E_a \Gamma_p)^2 \quad (4)$$

$$\bar{F}^* = \bar{F}_s/B \quad (5)$$

$$T_0^* = RT_0/E_a \quad (6)$$

We use the expression "simple thermal ignition" hereafter to denote the process in which the propellant surface is heated only by heat flux supplied externally and heat flux produced by the key ignition reaction. The more complicated ignition process in which secondary ignition reactions at or near the surface supply part of the heat flux will be described later. For simple thermal ignition,  $t_i$  is a function only of  $\bar{F}_s$  and  $T_0$  through propellant kinetic, thermophysical, and thermochemical parameters.

Baer found, by examining the results of his analysis, that, for simple thermal ignition, a class of propellants for which  $T_0^*$  is the same will conform to a unique  $t_i^*$  vs  $\bar{F}^*$  relationship. Furthermore, for experimental data, a plot of  $\bar{F}_s$  vs  $(t_i)^{1/2}$  on logarithmic coordinates should give a line having a slope of nearly minus one. For the range of parameters of interest, Baer found that the slope of line representing data plotted in this form is related to the activation energy of the key ignition reaction to a close approximation by

$$d \ln(t_i)^{1/2} / d \ln(\bar{F}_s) = 4.2(RT_0/E_a) - 1.0 \quad (7)$$

## Experimental Procedure

Ignition of propellants by convective heating was studied in a shock-tube apparatus similar to that used for earlier work at the University of Utah.<sup>6,7,11</sup> The propellant samples were mounted flush with a wall of a constant-area flow channel in a test section at the end of the shock tube and subjected to convective heating as the hot gas behind the reflected shock wave was released to the atmosphere through the channel. Figure 1 is a cutaway sketch of the shock tube, showing the position of the test section in the driven end. The test section shown by the sketch of Fig. 2 had a constant-area flow channel 0.500 in. wide by 0.125 in. high with a bell-mouthed entrance. The velocity of test gas through the channel was controlled by critical-flow orifices at the downstream end of the channel.

### Heat Transfer to Wall of Test Section

Heat transfer to the wall of the test-section channel at the test position was measured with heat-flux gages (thin-

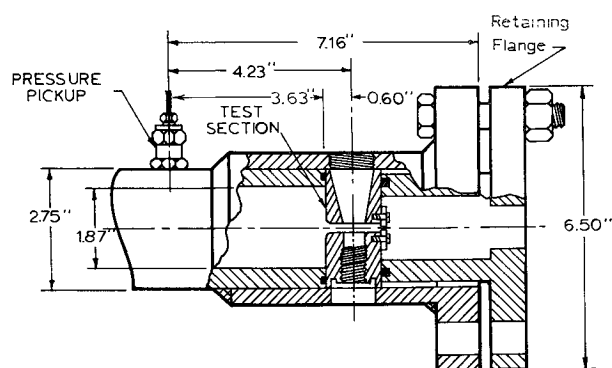


Fig. 1 Cross-sectional drawing of shock-tube driven end, showing the location of test section used for convective heating experiments.

film platinum resistance thermometers bonded to substrates of Pyrex or ceramic) that were interchangeable with the propellant sample holder. This heat-transfer study confirmed the earlier work of Baer<sup>6,7</sup> that the transient surface-temperature rise at the test position could be represented by the following equation:

$$(T_s - T_i)/(T_g - T_i) = (1 - e^{N^2} \operatorname{erfc} N) \quad (8)$$

$$N = h(t)^{1/2}/\Gamma \quad (9)$$

For temperatures defined on an absolute temperature scale,

$$T_i = T_0 + \Delta T_0(^{\circ}\text{K}) \quad (10)$$

The term  $\Delta T_0$  is a function of 1) Mach number of the incident shock wave, 2) pressure behind the reflected shock wave, 3) thermal responsivity of the gage substrate, and 4) area of the flow-control orifice downstream of the test position.

For the test section illustrated in Fig. 2, the heat-transfer coefficient  $h$  for heat transfer to the wall from the hot gas at the test position was correlated in terms of gas stagnation temperature  $T_g$ , and mass flow rate through the test-section channel  $G$ . Experimental heat-transfer coefficients were represented within an average deviation of 6% by the following equations:

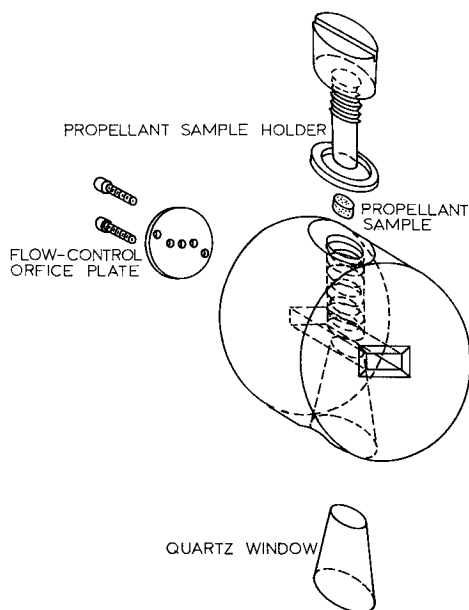


Fig. 2 Sketch of test section, showing propellant sample holder and critical-flow orifice plate.

For air, oxygen, and nitrogen,

$$h = 1.435 \times 10^{-4} (T_g)^{0.3} (G)^{0.905} \quad \text{cal/cm}^2 \text{ sec } ^{\circ}\text{K} \quad (11)$$

For argon,

$$h = 4.05 \times 10^{-4} (G)^{1.06} \quad \text{cal/cm}^2 \text{ sec } ^{\circ}\text{K} \quad (12)$$

where  $T_g$  is in  $^{\circ}\text{K}$  and  $G$  has the units  $\text{g/cm}^2 \text{ sec}$ . The larger dependence of  $h$  on  $G$  than observed for correlations of turbulent flow heat-transfer coefficients is attributed to early transition from laminar to turbulent flow in the boundary layer produced by high-intensity freestream turbulence. The information from this study was applied directly for calculating heat transfer to propellants during ignition tests, as described in the following section.

### Heat Transfer to Propellants

As mentioned earlier, data for simple thermal ignition of a propellant can be represented by a straight line if data are plotted in the form of  $\ln(\bar{F}_s)$  vs  $\ln(t_i)^{1/2}$ . The mean heat flux  $\bar{F}_s$  to the propellant surface during ignition tests is calculated, assuming that the propellant undergoes heating without reaction to a surface temperature  $T_{si}^L$ , up to the instant that ignition is observed, by the following equation:

$$\bar{F}_s = (\Gamma_p/2)(\Pi/t_i)^{1/2}(T_{si}^L - T_0) \quad (13)$$

$T_{si}^L$  is the "ignition temperature for linear heating"  $^{\circ}\text{K}$ , and is calculated with the following modification of Eq. (8):

$$T_{si}^L = T_i + (T_g - T_i)(1 - e^{N^2} \operatorname{erfc} N) \quad (14)$$

$$N = h(t_i)^{1/2}/\Gamma_p \quad (15)$$

The significance of  $T_{si}^L$  in the thermal ignition process is illustrated by the surface temperature-time relationship for a propellant undergoing simple thermal ignition in Fig. 3. Experimentally, the propellant surface was heated continuously by applied heat flux until a combustion flame was observed. The first appearance of this combustion flame at the surface, as observed by an RCA 1 P40 gas photodiode, was used as our criterion for ignition, as indicated on Fig. 3.

### Test Variables for Ignition

The externally applied heat flux to the propellant surface in convective heating experiments is dependent on the following parameters for a given test gas,

$$\bar{F}_s = f(T_g, U, P) \quad (16)$$

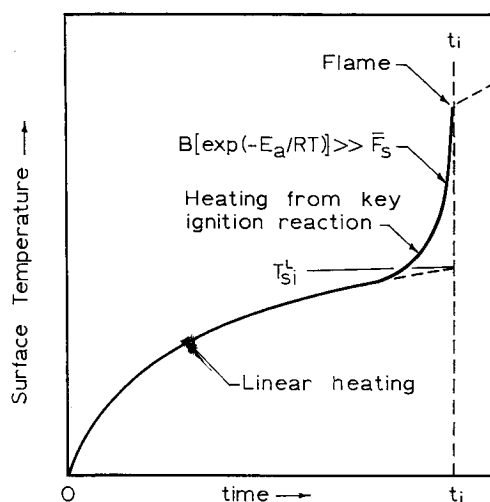


Fig. 3 Idealized surface-temperature history for a semi-infinite slab of propellant undergoing thermal ignition.

where  $T_g$ ,  $U$ , and  $P$  are the temperature, velocity, and pressure, respectively, of the test gas. When, as frequently is the case, ignition occurs by processes more complicated than simple thermal ignition, the ignition time  $t_i$  for a given propellant may be a function of one or more of the parameters  $T_g$ ,  $u$ , and  $P$ , as well as  $\bar{F}_s$ . The shock-tube technique does not allow the four parameters of Eq. (16) to be varied independently over significant ranges of the parameter values for a given test gas. Some separation of the effects was obtained by employing argon in place of nitrogen as an inert heating gas in some of the tests.

The range of values for test variables employed in ignition experiments were: 1) Heat flux: the heat flux  $\bar{F}_s$  at the propellant surface was varied by altering either the initial shock-tube conditions or the gas velocity through the constant-area flow channel of the test section; 2) Gas velocity: the velocity  $U$  of the test gas through the flow channel was varied over the range of 50 to 800 m/sec, corresponding to Mach numbers of 0.07 to 1.0, respectively; 3) Gas temperature: the temperature of the test gas  $T_g$  was varied over the range of 1000° to 2600°K°; and 4) Test pressure: propellant ignition was studied at pressures in the range of 14 to 25 atm.

### Ignition of Cast Propellants

A rather comprehensive study was made of the ignition characteristics of cast ammonium perchlorate (AP) propellants with a binder-fuel composed of a copolymer of polybutadiene and acrylic acid (PBAA) cured with an epoxy curing agent (Epon 828). Compositional data on the propellants discussed in this paper are given in Table 1, and their thermal properties are listed in Table 2.

#### Ignition of Propellant with Coarse Ammonium Perchlorate in Nitrogen

An extensive study was made of the ignition of  $F$ -propellant (see Table 1) by convective heating for the purpose of comparing ignition data with those obtained earlier at low radiant heat fluxes.<sup>8</sup> Samples for ignition tests were prepared by casting freshly mixed propellant directly into the cavities of the sample holders. The sample holders used were the same as that shown in Fig. 2. The cavity was overfilled so that a fresh surface could be cut immediately before ignition tests with a sharp razor blade. It was not possible to cut perfectly smooth surfaces on samples of  $F$ -propellant. It is estimated that a surface roughness of the order of 20 to 30  $\mu$  was produced by fracturing large crystals of AP and pulling some crystals from the propellant matrix.

Table 1 Propellant compositions

| Propellant code | Ingredients, wt %      |                       |                      |                                      |
|-----------------|------------------------|-----------------------|----------------------|--------------------------------------|
|                 | Fuel                   | Catalyst <sup>a</sup> | Ammonium perchlorate | AP particle size, $\mu$ <sup>b</sup> |
| CB              | 16.0 Philblack         | 2.0 Copper chromite   | 41.0                 | 15                                   |
|                 | E                      |                       | 41.0                 | 200                                  |
| F               | 18.0 PBAA <sup>c</sup> | 2.0 Copper chromite   | 40.0                 | 15                                   |
|                 |                        |                       | 40.0                 | 200                                  |
| G               | 18.0 PBAA <sup>c</sup> | None                  | 41.0                 | 15                                   |
|                 |                        |                       | 41.0                 | 200                                  |
| U               | 23.0 PBAA <sup>c</sup> | 2.0 Copper chromite   | 75.0                 | 15                                   |

<sup>a</sup> The copper chromite was catalyst Cu-0202 P from Harshaw Chemical Company.

<sup>b</sup> Fifty weight percent of the AP particles have diameters less than the value indicated.

<sup>c</sup> The PBAA binder-fuel was prepared from 85.0 weight percent liquid polybutadiene-acrylic acid copolymer and 15.0 weight percent Epon 828. Propellants were cured for one week at 80°C.

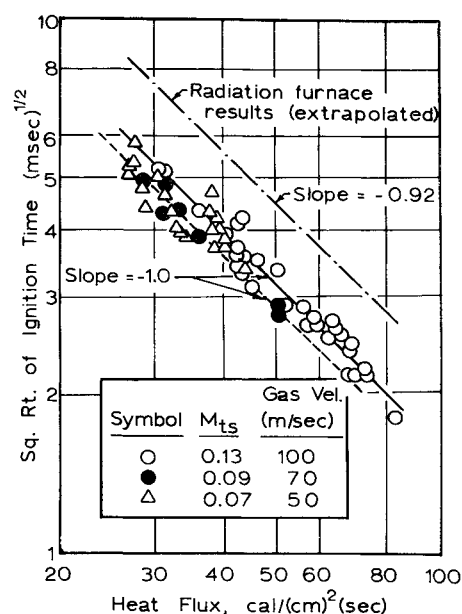


Fig. 4 Data for ignition of  $F$ -propellant by convective heating with nitrogen at low test-gas velocities for pressures in the range of 14 to 25 atm.

#### Ignition at low gas velocities

For the purpose of this discussion, low gas velocities are velocities of the convective gas through the test-section channel of 100 m/sec or less, corresponding to Mach numbers less than 0.15. The test-gas Mach number  $M_{ts}$  was determined by the size of the flow-control orifice used. It thus is convenient to present experimental data on propellant ignition as a plot of  $\ln(\bar{F}_s)$  vs  $\ln(t_i)^{1/2}$  for constant values of test-gas Mach number (or flow-control orifice size).

The ignition data on  $F$ -propellant at low gas velocities are shown in Fig. 4. (All of the data used in the preparation of figures for this paper are tabulated in Ref. 12.) The uppermost straight line in this figure is an extrapolation of the straight line that represented ignition data on  $F$ -propellant obtained in the radiation furnace at heat fluxes of 1–12 cal/cm<sup>2</sup> sec.<sup>8</sup> The ignition data given by Fig. 4 for convective heating all lie below the straight line defined by radiation-furnace results at 1 atm pressure in air.

Except for the rather large difference in ignition times for equivalent heat fluxes (extrapolated from lower fluxes) in radiant-heating and convective-heating experiments, the only other significant trend observed was that  $F$ -propellant was more difficult to ignite ( $t_i$  greater for given  $\bar{F}_s$ ) as the gas velocity across the sample surface was increased. The effect of gas velocity on ignition was expected, for it was observed previously by McCune<sup>11</sup> and more recently by Kling, Maman, and Brulard.<sup>13</sup>

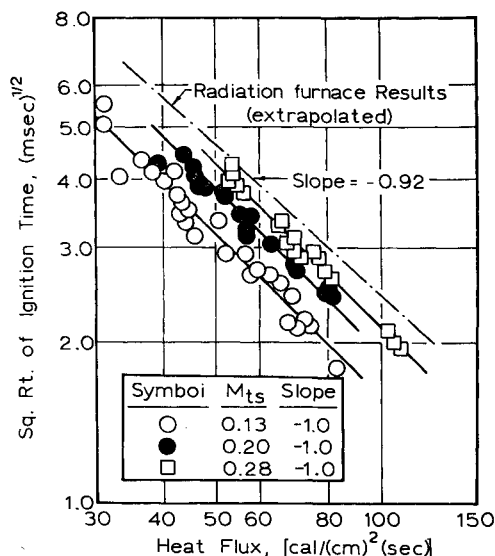
It was interesting to note that ignition data for the convective-heating experiments, when plotted in the form of

Table 2 Propellant thermophysical properties<sup>a</sup>

| Propellant code | Thermal diffusivity, cm <sup>2</sup> /sec | Heat capacity, cal/g °C | Density at 25°C, g/cm <sup>3</sup> | Thermal responsivity, cal/cm <sup>2</sup> sec <sup>1/2</sup> °C |
|-----------------|-------------------------------------------|-------------------------|------------------------------------|-----------------------------------------------------------------|
| CB              | 0.0024                                    | 0.260                   | 1.65                               | 0.021                                                           |
| F               | 0.00170                                   | 0.316                   | 1.63                               | 0.0212                                                          |
| G               | 0.00171                                   | 0.311                   | 1.60                               | 0.0206                                                          |
| U               | ...                                       | ...                     | 1.58                               | 0.0203 <sup>b</sup>                                             |

<sup>a</sup> Propellant properties are for a temperature of approximately 60°C.

<sup>b</sup> Thermal responsivity was estimated for this propellant from experimental data on a similar propellant.



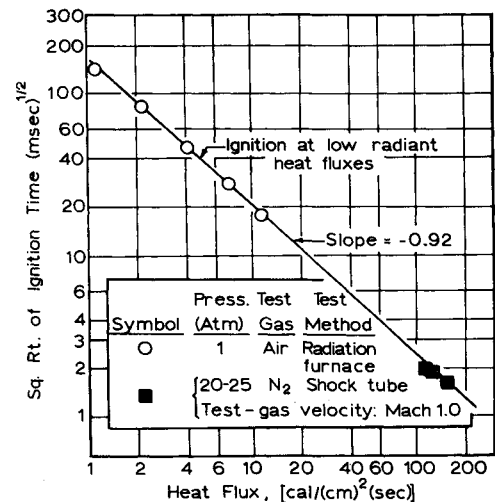
**Fig. 5** Data for ignition of *F*-propellant by convective heating with nitrogen at intermediate test-gas velocities, 130 to 235 m/sec, for pressures in the range of 20 to 25 atm.

$\ln(\bar{F}_s)$  vs  $\ln(t_i)^{1/2}$ , were well represented by a straight line with a slope of about  $-1.0$ . The ignition data obtained in the radiation furnace were represented by a straight line with a slope of  $-0.92$ .

**Ignition at intermediate gas velocities**

In the next set of experiments on *F*-propellant, the velocity of the gas was increased to Mach 0.20 and Mach 0.28, corresponding to gas velocities of about 150 and 200 m/sec, respectively. Data obtained are presented graphically in Fig. 5. The effect of higher gas velocities was to increase further the time for ignition to occur at equivalent heat fluxes over that observed at lower gas velocities. Again, ignition data were represented by a straight line on the logarithmic plot of  $\bar{F}_s$  vs  $(t_i)^{1/2}$  with a slope of about  $-1.0$ .

The fact that gas velocity had such a large effect on ignition suggested that gas-phase or heterogeneous-surface reactions were important in the ignition process; so, as the gas velocity was increased, reactive species at or near the surface were being diluted and swept away by the fast-moving



**Fig. 6** Data for ignition of *F*-propellant by convective heating with nitrogen at high test-gas velocities, 720 to 805 m/sec, for pressures of 20 to 25 atm compared with data for ignition of *F*-propellant at low radiant heat fluxes in a radiation furnace at atmospheric pressure.

gas. Such an effect on the ignition process would be expected if gas-phase species were important in the ignition process.

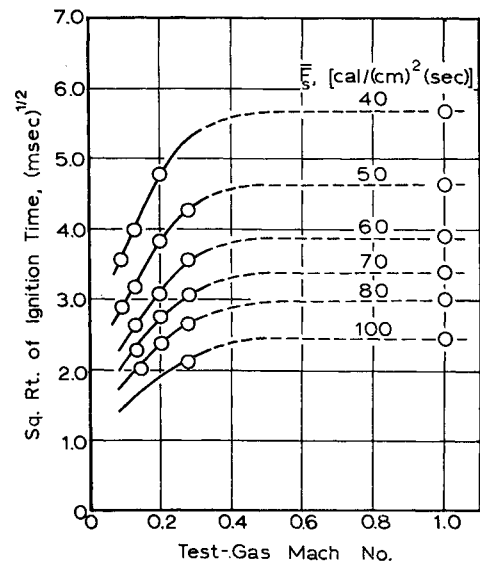
**Ignition at high gas velocities**

As a final test of the effect of gas velocity on ignition, runs were made on *F*-propellant at gas velocities of about 700 to 800 m/sec, which corresponded to a Mach number of about 1.0. If gas-phase or heterogeneous reactions were essential processes in ignition, it was expected that the *F*-propellant might not ignite when the test-gas velocity was equal to Mach 1.0. *F*-propellant ignited at Mach 1.0; however, steady burning of the propellant was extinguished later when cold driver gas entered the flow channel. When the ignition data obtained at Mach 1.0 were graphed in the form of  $\ln(\bar{F}_s)$  vs  $\ln(t_i)^{1/2}$ , the data were well represented by the extrapolated straight line for ignition results in the radiation furnace (Fig. 6).

These results at Mach 1.0 were unexpected, for they suggested that *F*-propellant undergoes simple thermal ignition only at high gas velocities and that, at lower gas velocities ignition was accelerated by a supplementary mechanism. The data for all velocities also show, when cross-plotted as in Fig. 7, that at higher gas velocities the effect of gas velocity on ignition time is diminished.

**Discussion of Velocity Effect on Ignition**

The explanation of the velocity effect requires that we anticipate the results of igniting propellant samples having smoother surfaces. As will be shown, the effect of velocity is much diminished for these propellants; and results for all velocities are well represented by the  $t_i$  vs  $\bar{F}_s$  relationship for low-flux radiative and high-velocity convective heating of *F*-propellant. In summary, at low heat flux levels (1 to 12 cal/cm<sup>2</sup> sec), surface roughness had no effect; but the effect of gas velocity was not investigated (radiation through a stagnant gas); at high heat flux levels (80 to 150 cal/cm<sup>2</sup> sec), the extrapolated heat flux-ignition time relationship obtained at low heat fluxes was confirmed when either the velocity of the heating gas was high (approaching Mach 1) or the propellant surface was smooth. Reduced ignition times at a given heat flux were observed when both surfaces were rougher and gas velocities were lower (less than Mach 0.5).



**Fig. 7** Effect of test-gas velocity on ignition time for convective heating of *F*-propellant with nitrogen for different values of externally applied heat flux; (this graph is a cross-plot of the results given by Figs. 4-6).

Our explanation is as follows. The surface irregularities, primarily the projections at the surface, which have large surface-to-volume ratios, are heated more rapidly than the propellant bounded by a flat surface. When the projections are small, relative to the depth of penetration of the thermal wave, the heat flux at the surface can be calculated by one-dimensional equations. When, on the other hand, the projections are relatively large (20 to 40  $\mu$ ), the temperature of the projections quickly becomes higher than the one-dimensional values that induce early decomposition. The products of decomposition react exothermically with each other in the gas phase or heterogeneously at the propellant surface, thus giving energy to augment the external flux and reduce the ignition time. (Reactions of this kind are referred to as "secondary ignition reactions" in this paper.) If the velocity of the heating gas is increased, the decomposition products are swept into the heating gas before reacting so that the heat flux-augmenting effect of the decomposing projections is reduced.

This picture of simple thermal ignition supplemented under some conditions by secondary ignition reactions is illustrated by the idealized surface temperature-time relationships of Fig. 8. Curve 1 represents the surface-temperature history for a propellant undergoing simple thermal ignition by linear heating from externally applied heat flux only to the time of ignition; no augmentation of the external heat flux by secondary ignition reactions is shown. At  $(t_i)_1$  for simple thermal ignition (Fig. 8), "the ignition temperature for linear heating,"  $T_{si}^L$ , has been defined as the "thermal-ignition temperature,"  $T_{si}^T$ . The thermal-ignition temperature is a property of the propellant and is the temperature to which the propellant surface must be raised before the propellant undergoes thermal ignition. Curve 2 of Fig. 8 represents the surface-temperature history for a propellant undergoing thermal ignition with external heat flux supplemented by secondary ignition reactions. It is seen that the propellant surface is raised to its thermal-ignition temperature more rapidly in time  $(t_i)_2$  with the aid of energy from secondary ignition reactions.

As we see the ignition process, the heat flux  $\bar{F}_{sr}$  contribution from secondary ignition reactions heats the propellant surface in the same way as the externally applied heat flux, i.e., by linear heating; however, we cannot, for the present,

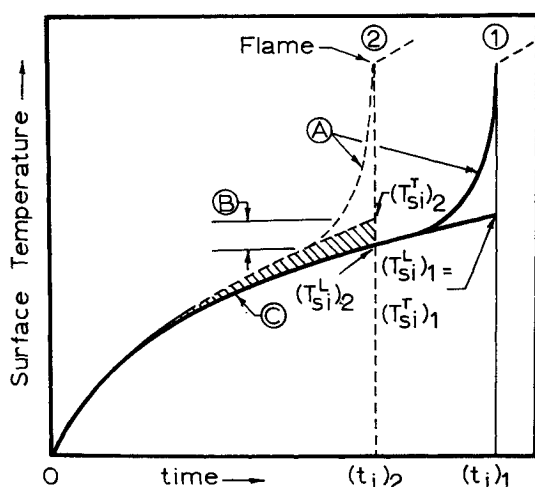


Fig. 8 Surface-temperature history, for a propellant undergoing thermal ignition, illustrating the effect of secondary ignition reactions on the ignition process; curve 1 is for simple thermal ignition; curve 2 is for thermal ignition with augmentation of the externally applied heat flux by exothermic secondary ignition reactions; A is surface heating from key ignition reaction, B is surface heating from secondary ignition reactions, and C is linear heating from externally applied heat flux.

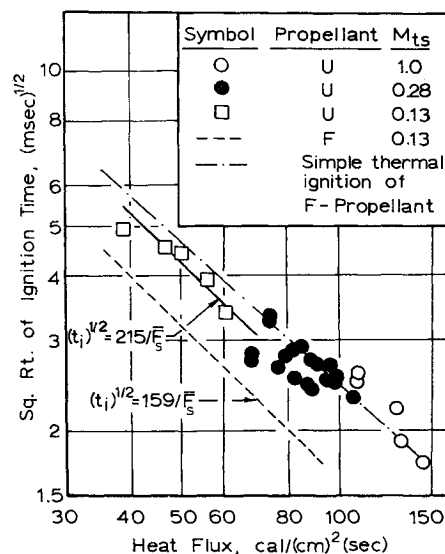


Fig. 9 Data for ignition of U-propellant by convective heating with nitrogen for pressures of 20 to 25 atm.

explicitly calculate this contribution to the heating process. Therefore, we have restricted our definition of  $T_{si}^L$  to the temperature rise at the propellant surface produced by the externally applied heat flux only. The combined effect of the two processes, when secondary ignition reactions contribute heat flux, brings the propellant to its thermal-ignition temperature  $T_{si}^T$  in the experimentally observed ignition time.

The  $T_{si}^T$  for F-propellant can be expressed in terms of either the externally applied heat flux or the experimentally observed ignition time by the following equations, for an initial uniform propellant temperature of 300°K:

$$T_{si}^T(^{\circ}\text{K}) = 300^{\circ}\text{K} + 286.1 (\bar{F}_s + \bar{F}_{sr})^{0.08} \quad (17)$$

$$T_{si}^T(^{\circ}\text{K}) = 300^{\circ}\text{K} + 443.4 (t_i)^{-0.9435} \quad (18)$$

where  $t_i$  is in milliseconds and  $\bar{F}_s$  has the units  $\text{cal}/\text{cm}^2 \text{ sec}$ . For simple thermal ignition, the parameter  $\bar{F}_{sr}$  in Eq. (17) is zero.

#### Ignition of Propellant with Fine Ammonium Perchlorate in Nitrogen

For studying the ignition of propellants with fine particle-size AP, U-propellant containing a single cut of 15  $\mu$  (average diameter) AP and 2% of copper chromite was used (see Table 1). The particle-size distribution in the 15  $\mu$  cut of AP varied from diameters of less than 1 to 25  $\mu$ . It was necessary to reduce the loading of AP from 80% in F-propellant to 75% in U-propellant to obtain a castable material. It had been established previously<sup>14</sup> that the loading of ammonium perchlorate had no appreciable effect on ignition characteristics for our test conditions. Carefully cut surfaces on this propellant are estimated to have a surface roughness of about 5  $\mu$ .

Ignition data for U-propellant in nitrogen for test-gas velocities of Mach 0.13, 0.28, and 1.0 are compared to ignition results for F-propellant in Fig. 9. The equations in Fig. 9 are for the straight lines shown, where  $\bar{F}_s$  has the units of  $\text{cal}/\text{cm}^2 \text{ sec}$  and  $t_i$  is in milliseconds. A significant feature of these data is that they are all grouped near the straight line that represents simple thermal ignition of F-propellant, i.e., ignition data for F-propellant obtained at low radiant heat fluxes and data for convective heating at high gas velocities. Thus, for propellants with a smooth surface, the effect of gas velocity on ignition almost is negligible.

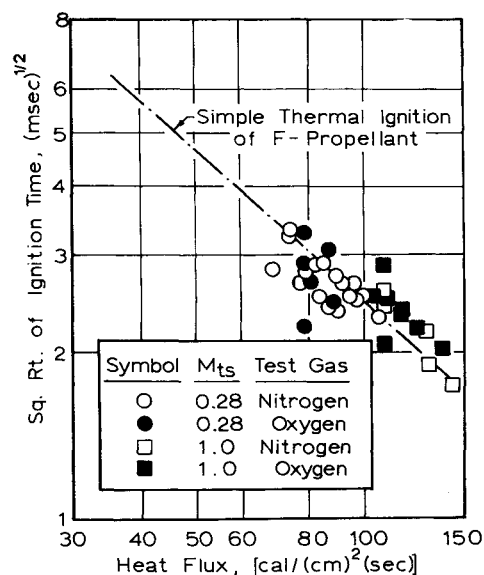


Fig. 10 Data for ignition of *U*-propellant by convective heating with oxygen for test-gas velocities of Mach 0.28 and 1.0 for pressures of 20 to 25 atm.

#### Effect of Oxidizing Species in Test Gas

Other investigators<sup>6,7,11,13,15</sup> have observed that the presence of oxygen in the test gas gives a rather significant improvement in the ignitability of composite propellants containing ammonium perchlorate. Since it was expected that pure oxygen greatly would affect the ignition of *F*-propellant, this propellant was tested only in air. *U*-propellant was tested in pure oxygen.

Ignition data for *U*-propellant in pure oxygen are compared with results for ignition in nitrogen at Mach 0.28 and Mach 1.0 in Fig. 10. It was found that *F*-propellant ignited faster in air than in nitrogen at equivalent heat fluxes; on the other hand, samples of *U*-propellant with smooth surfaces exhibited the same ignition characteristics in both oxygen and nitrogen. These results provided further evidence that only for propellant samples with rough surfaces do secondary ignition processes complicate the thermal-ignition process. They also suggest that oxidizing species in the test gas affect ignition by participating in the secondary ignition process previously described. The experimental results also indicate that heterogeneous attack at the propellant surface by molecular oxygen is not an important ignition reaction in the range of conditions studied.

#### Ignition of Pressed Propellants

As a means of obtaining more information on the role of propellant ingredients in ignition, pressed propellant *CB* with carbon black as the fuel was tested in oxygen and nitrogen at pressures of 20 to 25 atm. Ignition data for samples of propellant containing carbon black, with smooth surfaces, in nitrogen and oxygen at Mach 0.28 are compared with data obtained in the radiation furnace at low radiant heat fluxes in Fig. 11. The data for convective heating again are in good agreement with the straight line that was extrapolated from radiation-furnace results at atmospheric pressure in air.<sup>8</sup> It was found that samples of propellant *CB* with surface irregularities ignited faster in convective-heating experiments than those with smooth surfaces. This occurrence suggests that the secondary ignition reactions at the surface are primarily reactions among AP decomposition products. It would not be expected that the rather unreactive carbon black would undergo vigorous reactions at the low temperatures of the propellant surface during transient heating.

#### Conclusions

The term "propellant ignition" denotes an enormously complex process in which a number of subprocesses occur. The relative importance of these subprocesses is determined by the manner in which any of several effective external stimuli act upon the very complicated and different chemical systems that are propellants. The firmest conclusion that evolves from the intensive research of recent years is that there cannot be a single, coherent, valid theory of ignition which describes all instances of ignition in terms of a single rate-determining mechanism. Limiting cases, however, can be discerned as applying under a large range of conditions to a class of propellants.

A case applying to catalyzed ammonium perchlorate propellants is described here. We observe, under some conditions, a simple thermal ignition process, in which the propellant surface is heated without reaction to a temperature level at which a key ignition reaction involving the solid phases occurs. Products of this reaction then react rapidly to complete the ignition process. The quantitative description by Baer<sup>8</sup> fits this observation well.

Under other conditions, what we call secondary ignition reactions accelerate the process. Apparently, they are reactions involving gaseous products of decomposition evolved at local hot spots on the surface. Their effect can be eliminated either by eliminating the hot spots, as by employing smooth surfaces, or by removing the gaseous reactants, as by increasing the gas velocity across the surface. When these measures are taken, the ignition time-heat flux relationship described by simple thermal ignition theory is observed again.

For the experimental data discussed here, the agreement with simple thermal ignition theory is too good to be fortuitous. Although first approximation agreement would be anticipated because most of the ignition time is used for heating the propellant, the precision of both the low heat flux and high heat flux data is good enough to obtain a numerical value for the slope of the line that represents data plotted in the form of  $\ln(\bar{F}_s)$  vs  $\ln(t_i)^{1/2}$ . This numerical value for the slope has the significance that it is related to the activation energy of the slow step in the ignition process, a key reaction the rate of which depends on the temperature of the solid surface.

The following facts lead us to postulate that the key ignition reaction is the low-temperature thermal decomposition of ammonium perchlorate:

1) Different binder-fuels give the same ignition results when ignition data are plotted in the dimensionless form

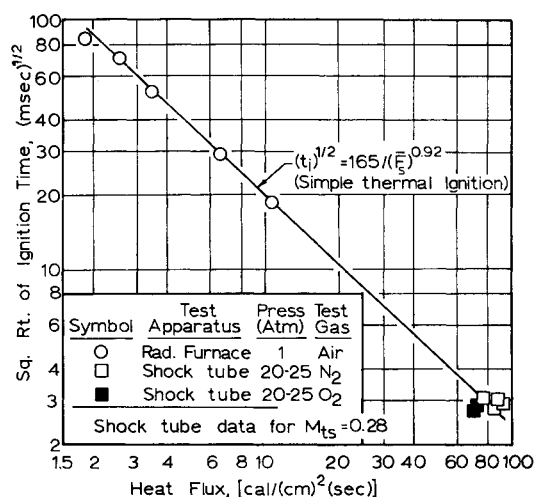


Fig. 11 Data for ignition of pressed propellant *CB* (with carbon black fuel) by high convective heat fluxes in nitrogen and oxygen and by low radiant heat fluxes in air.

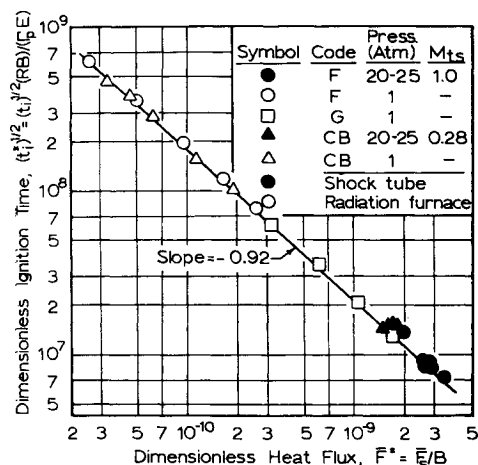


Fig. 12 Ignition data for simple thermal ignition of cast propellants *F* and *G* and pressed propellant *CB*, plotted in dimensionless form, as suggested by thermal ignition theory.

suggested by theory (Fig. 12). Data for simple thermal ignition of three propellants, propellant *G* (uncatalyzed), propellant *F* (catalyzed with 2% copper chromite) (both with PBAA binder-fuel), and propellant *CB* with carbon black as the fuel (catalyzed with 2% copper chromite) are represented by one straight line when plotted in this form. Complete agreement is found with the low heat flux results of previous studies<sup>8</sup> in which polysulfide, polyurethane, and methylvinylpyridine-butadiene polymers, and graphite were used as fuels.

2) The only other oxidizer used in this laboratory, ammonium nitrate,<sup>8</sup> does not give agreement.

3) The activation energy  $E_a$  computed from Eq. (7) for the key ignition reaction, approximately 30 kcal/mole, is in good agreement with that observed for the low-temperature decomposition of ammonium perchlorate by other workers.<sup>9,16-18</sup>

For the conditions of these experiments, high pressure and high test-gas temperature, the rate of AP decomposition determines the rates of all subsequent reactions. The rates of these later reactions, regardless of whether they occur in the gas phase or at the solid surface, apparently are rapid enough that they do not influence the time required for the over-all process. If we alter the conditions of the experiments by lowering the pressure or by permitting cold gas in contact with propellant, it appears likely that reactions other than the decomposition of AP may become the slow step in the ignition process.

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